

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler
```

```
In [4]: df = pd.read_csv("C:/Users/pratiksha/Downloads/archive (2)/titanic-Dataset.csv")
df.head()
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarke
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	

```
In [5]: df.tail(5)
```

Out[5]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [7]: df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [8]: df.isnull()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	False	False	False	False	False	False	False	False	False	False	True	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	True	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	True	False
...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	True	False
887	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	False	False	True	False	False	False	False	True	False
889	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	True	False

891 rows × 12 columns

```
In [9]: df.isnull().sum()
```

```
Out[9]: PassengerId    0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                 177
SibSp                0
Parch                0
Ticket              0
Fare                 0
Cabin               687
Embarked             2
dtype: int64
```

```
In [10]: df.notnull()
```

Out[10]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	True	True	True	True	True	True	True	True	True	True	False	True
1	True	True	True	True	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True	True	True	False	True
3	True	True	True	True	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True	True	True	False	True
...	...	...	...	...	...	...	...	...	...	...	...	...
886	True	True	True	True	True	True	True	True	True	True	False	True
887	True	True	True	True	True	True	True	True	True	True	True	True
888	True	True	True	True	True	False	True	True	True	True	False	True
889	True	True	True	True	True	True	True	True	True	True	True	True
890	True	True	True	True	True	True	True	True	True	True	False	True

891 rows × 12 columns

```
In [11]: df.notnull().sum()
```

```
Out[11]: PassengerId    891
Survived              891
Pclass               891
Name                 891
Sex                  891
Age                 714
SibSp                891
Parch                891
Ticket              891
Fare                 891
Cabin               204
Embarked            889
dtype: int64
```

```
In [12]: df.shape
```

```
Out[12]: (891, 12)
```

```
In [13]: df.dtypes
```

```
Out[13]: PassengerId    int64
         Survived      int64
         Pclass        int64
         Name          object
         Sex           object
         Age           float64
         SibSp         int64
         Parch         int64
         Ticket        object
         Fare          float64
         Cabin         object
         Embarked      object
         dtype: object
```

```
In [14]: df['Age'] = df['Age'].astype('float')
         df['Fare'] = df['Fare'].astype('float')
```

```
In [15]: df['Age'] = df['Age'].fillna(df['Age'].mean())
         df
```

Out[15]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.000000	1	0	PC 17599	71.2833	C85	S
2	3	1	3	Heikkinen, Miss. Laina	female	26.000000	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.000000	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	29.699118	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.000000	0	0	111369	30.0000	C148	S
890	891	0	3	Dooley, Mr. Patrick	male	32.000000	0	0	370376	7.7500	NaN	S

891 rows × 12 columns

In [18]:

```
df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
Out[18]: 0      S
          1      C
          2      S
          3      S
          4      S
          ..
        886      S
        887      S
        888      S
        889      C
        890      Q
        Name: Embarked, Length: 891, dtype: object
```

```
In [19]: df.drop(columns=['Cabin'], inplace=True)
```

```
In [21]: print(df.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64
```

```
In [22]: scaler = MinMaxScaler()
df[['Age', 'Fare']] = scaler.fit_transform(df[['Age', 'Fare']])
```

```
In [23]: print(df[['Age', 'Fare']].head())
```

```
      Age      Fare
0  0.271174  0.014151
1  0.472229  0.139136
2  0.321438  0.015469
3  0.434531  0.103644
4  0.434531  0.015713
```

```
In [24]: df['Sex'] = df['Sex'].map({'male': 1, 'female': 0})
```

```
In [25]: df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)
```

```
In [28]: df[['Embarked_Q', 'Embarked_S']] = df[['Embarked_Q', 'Embarked_S']].astype(int)
```

```
In [29]: print(df.dtypes)
```

```
PassengerId    int64
Survived        int64
Pclass          int64
Name            object
Sex             int64
Age            float64
SibSp           int64
Parch           int64
Ticket          object
Fare            float64
Embarked_Q      int64
Embarked_S      int64
dtype: object
```

```
In [30]: print(df.head())
```

```

PassengerId  Survived  Pclass  \
0            1          0        3
1            2          1        1
2            3          1        3
3            4          1        1
4            5          0        3

```

```

                                Name  Sex      Age  SibSp  \
0                        Braund, Mr. Owen Harris    1  0.271174    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...    0  0.472229    1
2                        Heikkinen, Miss. Laina    0  0.321438    0
3      Futrelle, Mrs. Jacques Heath (Lily May Peel)    0  0.434531    1
4                        Allen, Mr. William Henry    1  0.434531    0

```

```

Parch      Ticket      Fare  Embarked_Q  Embarked_S
0      0      A/5 21171  0.014151         0         1
1      0      PC 17599  0.139136         0         0
2      0  STON/O2. 3101282  0.015469         0         1
3      0      113803  0.103644         0         1
4      0      373450  0.015713         0         1

```

In []: